

TIRA E-COMMERCE DATABASE MANAGEMENT SYSTEM

Seller Inventory Management System

1.1 Seller Table

Attribute	Data Type	Constraint
Seller_ID	INT	Primary key
Seller_Name	VARCHAR(100)	NOT NULL
Email	VARCHAR(100)	NOT NULL,UNIQUE
Contact_No	VARCHAR(15)	NOT NULL,UNIQUE
Address	VARCHAR(200)	NOT NULL

1.2 Inventory Table

Attribute	Data Type	Constraint
Inventory_ID	INT	Primary Key
Product_ID	INT	Foreign Key
Seller_ID	INT	Foreign Key
Stock	INT	NOT NULL
Status	Varchar(20)	NOT NULL

2. Table Relationship

Seller → Inventory

- Seller.Seller_ID → Primary Key
- Inventory.Inventory_ID → Primary Key
- Inventory.Seller_ID → Foreign Key referencing Seller.Seller_ID
- Relationship: One-to-Many (1:M)
- One seller can have multiple inventory records.

Product → Inventory

- Product.Product_ID → Primary Key
- Inventory.Inventory_ID → Primary Key
- Inventory.Product_ID → Foreign Key referencing Product.Product_ID
- Relationship: One-to-Many (1:M)
- One product can have multiple inventory records.

3. Table Creation

3.1 Create Seller Table

```
SQL> CREATE TABLE Seller (  
2     Seller_ID NUMBER PRIMARY KEY,  
3     Seller_Name VARCHAR2(100) NOT NULL,  
4     Phone VARCHAR2(15),  
5     Email VARCHAR2(100),  
6     Address VARCHAR2(200)  
7 );
```

Table created.

```
SQL>
```

3.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (  
2     Inventory_ID NUMBER PRIMARY KEY,  
3     Seller_ID NUMBER,  
4     Product_ID NUMBER,  
5     Stock NUMBER,  
6     Status VARCHAR2(20),  
7     FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID),  
8     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)  
9 );
```

Table created.

4 Seller Data Insertion Records

```
SQL> INSERT INTO Seller VALUES
  2 (201, 'Lakshmi Stores', '9876543210', 'lakshmi@gmail.com', 'Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (202, 'Beauty Hub', '9876543211', 'beautyhub@gmail.com', 'Coimbatore');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (203, 'Glow Mart', '9876543212', 'glowmart@gmail.com', 'Madurai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (204, 'Style World', '9876543213', 'styleworld@gmail.com', 'Salem');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 (205, 'Care Point', '9876543214', 'carepoint@gmail.com', 'Trichy');

1 row created.

SQL>
SQL> COMMIT;
```

5 Inventory Records

```
SQL> INSERT INTO Inventory VALUES
  2  (1, 201, 102, 60, 'Available');

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES
  2  (2, 202, 103, 30, 'Available');

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES
  2  (3, 203, 104, 25, 'Available');

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES
  2  (4, 204, 105, 0, 'Unavailable');

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES
  2  (5, 205, 102, 0, 'Unavailable');

1 row created.

SQL>
SQL> COMMIT;
```

6.Seller Product Information

6.1 Display Seller and Product Details

```
SQL> SELECT
  2     s.Seller_Name,
  3     p.Product_Name,
  4     i.Stock AS Stock_Quantity,
  5     i.Status AS Stock_Status
  6 FROM Seller s
  7 JOIN Inventory i
  8 ON s.Seller_ID = i.Seller_ID
  9 JOIN Product p
 10 ON i.Product_ID = p.Product_ID
 11 ORDER BY s.Seller_Name;
```

SELLER_NAME

PRODUCT_NAME

STOCK_QUANTITY STOCK_STATUS

Beauty Hub

Shampoo

30 Available

Care Point

Moisturizer

0 Unavailable

SELLER_NAME

PRODUCT_NAME

STOCK_QUANTITY STOCK_STATUS

Glow Mart

Lipstick

25 Available

Lakshmi Stores

Moisturizer

SELLER_NAME

7.Inventory Status

7.1 Display Available Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Stock AS Stock_Quantity,
  5     i.Status AS Stock_Status
  6 FROM Product p
  7 JOIN Inventory i
  8 ON p.Product_ID = i.Product_ID
  9 WHERE i.Status = 'Available';
```

PRODUCT_ID

PRODUCT_NAME

STOCK_QUANTITY STOCK_STATUS

102	
Moisturizer	
60	Available

103	
Shampoo	
30	Available

PRODUCT_ID

PRODUCT_NAME

STOCK_QUANTITY STOCK_STATUS

104	
Lipstick	
25	Available

7.2 Display Unavailable Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Stock AS Stock_Quantity,
  5     i.Status AS Stock_Status
  6 FROM Product p
  7 JOIN Inventory i
  8 ON p.Product_ID = i.Product_ID
  9 WHERE i.Status = 'Unavailable';
```

PRODUCT_ID

PRODUCT_NAME

STOCK_QUANTITY STOCK_STATUS

104

Lipstick

0 Unavailable

8.Inventory Update

```
SQL> UPDATE Inventory
  2 SET Stock = 30,
  3     Status = 'Available'
  4 WHERE Inventory_ID = 3;

1 row updated.
```

9.Inventory Status Reports

9.1 Complete Inventory Report

```
SQL> SELECT
  2     i.Inventory_ID,
  3     s.Seller_Name,
  4     p.Product_ID,
  5     p.Product_Name,
  6     i.Stock,
  7     i.Status
  8 FROM Inventory i
  9 JOIN Seller s
 10 ON i.Seller_ID = s.Seller_ID
 11 JOIN Product p
 12 ON i.Product_ID = p.Product_ID
 13 ORDER BY i.Inventory_ID;
```

INVENTORY_ID

SELLER_NAME

PRODUCT_ID

PRODUCT_NAME

STOCK STATUS

1

Lakshmi Stores

102

INVENTORY_ID

SELLER_NAME

PRODUCT_ID

PRODUCT_NAME

STOCK STATUS

Moisturizer

60 Available

INVENTORY_ID

SELLER_NAME

PRODUCT_ID

PRODUCT_NAME

STOCK STATUS

2

Beauty Hub

103

9.2 Available and Unavailable Product Count

```
SQL> SELECT
  2     Status AS Stock_Status,
  3     COUNT(*) AS Total_Products
  4 FROM Inventory
  5 GROUP BY Status;
```

STOCK_STATUS	TOTAL_PRODUCTS
Available	2
Unavailable	1

9.3 Total Stock By Seller

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     SUM(i.Stock) AS Total_Stock
  5 FROM Seller s
  6 JOIN Inventory i
  7 ON s.Seller_ID = i.Seller_ID
  8 GROUP BY s.Seller_ID, s.Seller_Name
  9 ORDER BY s.Seller_ID;
```

SELLER_ID	SELLER_NAME	TOTAL_STOCK
201	Lakshmi Stores	60
202	Beauty Hub	30

SELLER_ID	SELLER_NAME	TOTAL_STOCK
203	Glow Mart	0

9.4 Products With Zero Stock

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Stock,
  5     i.Status
  6 FROM Product p
  7 JOIN Inventory i
  8 ON p.Product_ID = i.Product_ID
  9 WHERE i.Stock = 0;
```

PRODUCT_ID

PRODUCT_NAME

STOCK STATUS

104

Lipstick

0 Unavailable